

Drop table dept Cascade Constraints;

-- Create the DEPT table

```
CREATE TABLE dept (  
    deptno NUMBER PRIMARY KEY,  
    dname VARCHAR2(50),  
    loc VARCHAR2(50)  
);
```

```
SQL> desc dept;  
Name                                     Null?      Type  
-----  
DEPTNO                                  NOT NULL   NUMBER  
DNAME                                   VARCHAR2(50)  
LOC                                     VARCHAR2(50)
```

```
SQL> select * from dept;  
  
no rows selected
```

Drop table emp Cascade Constraints;

-- Create the EMP table

```
CREATE TABLE emp (  
    empno NUMBER PRIMARY KEY,  
    ename VARCHAR2(50),  
    job VARCHAR2(30),  
    mgr NUMBER,  
    hiredate DATE,  
    sal NUMBER(8,2),  
    comm NUMBER(8,2),  
    deptno NUMBER,  
    FOREIGN KEY (deptno) REFERENCES dept(deptno)  
);
```

```
SQL> desc emp;
```

Name	Null?	Type
EMPNO	NOT NULL	NUMBER
ENAME		VARCHAR2(50)
JOB		VARCHAR2(30)
MGR		NUMBER
HIREDATE		DATE
SAL		NUMBER(8,2)
COMM		NUMBER(8,2)
DEPTNO		NUMBER

```
SQL> select * from emp;
```

```
no rows selected
```

```
-- populate data
```

```
-- Insert data into DEPT table
```

```
INSERT INTO dept (deptno, dname, loc) VALUES (1, 'ACCOUNTING', 'NEW YORK');
```

```
INSERT INTO dept (deptno, dname, loc) VALUES (2, 'RESEARCH', 'DALLAS');
```

```
INSERT INTO dept (deptno, dname, loc) VALUES (3, 'SALES', 'CHICAGO');
```

```
INSERT INTO dept (deptno, dname, loc) VALUES (7, 'IT', 'SAN FRANCISCO');
```

```
-- Insert data into EMP table
```

```
INSERT INTO emp (empno, ename, job, mgr, hiredate, sal, comm, deptno)
```

```
VALUES (101, 'SMITH', 'CLERK', NULL, TO_DATE('2023-01-15','YYYY-MM-DD'), 1200, NULL, 1);
```

```
INSERT INTO emp (empno, ename, job, mgr, hiredate, sal, comm, deptno)
```

```
VALUES (102, 'ALLEN', 'SALESMAN', 101, TO_DATE('2023-02-10','YYYY-MM-DD'), 1600, 300, 3);
```

```
INSERT INTO emp (empno, ename, job, mgr, hiredate, sal, comm, deptno)
```

```
VALUES (103, 'WARD', 'SALESMAN', 101, TO_DATE('2023-03-01','YYYY-MM-DD'), 1250, 500, 3);
```

```
INSERT INTO emp (empno, ename, job, mgr, hiredate, sal, comm, deptno)
```

```
VALUES (104, 'JONES', 'MANAGER', NULL, TO_DATE('2023-04-20','YYYY-MM-DD'), 2975, NULL, 2);
```

```
INSERT INTO emp (empno, ename, job, mgr, hiredate, sal, comm, deptno)
```

```
VALUES (105, 'MILLER', 'CLERK', 104, TO_DATE('2023-05-11','YYYY-MM-DD'), 1300, NULL, 2);
```

-- Employees in department 7

```
INSERT INTO emp (empno, ename, job, mgr, hiredate, sal, comm, deptno)
VALUES (106, 'DAVIS', 'ANALYST', 104, TO_DATE('2023-06-18','YYYY-MM-DD'), 3000, NULL, 7);
```

```
INSERT INTO emp (empno, ename, job, mgr, hiredate, sal, comm, deptno)
VALUES (107, 'ADAMS', 'DEVELOPER', 106, TO_DATE('2023-07-05','YYYY-MM-DD'), 2500, NULL,
7);
```

-- Create the VIEW for employees in department 7

```
SQL> CREATE VIEW emp_dept7 AS
SELECT empno, ename, sal
FROM emp
WHERE deptno = 7;
      2      3      4
View created.
```

```
SQL> SELECT * FROM emp_dept7;
```

EMPNO	ENAME	SAL
106	DAVIS	3000
107	ADAMS	2500

--Create Simple View Column Alias Example

```
SQL> CREATE VIEW emp_annualSalary
AS SELECT empno employeeID, ename employeeNAME,
sal*12 annualSALARY
FROM emp
WHERE deptno = 3;
      2      3      4      5
View created.
```

```
SQL> SELECT * FROM emp_annualSalary;
```

EMPLOYEEID	EMPLOYEENAME	ANNUALSALARY
102	ALLEN	19200
103	WARD	15000

- Rules 1 for Performing DML Operations on a View
- Can usually perform DML operations
- on simple views.

```
SQL> INSERT INTO emp_dept7 (empno, ename, sal) VALUES (9999, 'KO KO', 2000);

1 row created.

SQL> SELECT * FROM emp_dept7;
```

EMPNO	ENAME	SAL
106	DAVIS	3000
107	ADAMS	2500

- cannot insert BECAUSE WHERE CLAUSE

```
SQL> CREATE OR REPLACE VIEW emp_dept7 AS
SELECT empno, ename, sal, deptno
FROM emp
WHERE deptno = 7;
View created.

SQL> INSERT INTO emp_dept7 (empno, ename, sal, deptno) VALUES (4444, 'NYI NYI', 3000, 7);

1 row created.

SQL> SELECT * FROM emp_dept7;
```

EMPNO	ENAME	SAL
106	DAVIS	3000
107	ADAMS	2500
4444	NYI NYI	3000

```
SQL> DELETE FROM emp_dept7 WHERE empno=4444;
```

```
1 row deleted.
```

```
SQL> SELECT * FROM emp_dept7;
```

EMPNO	ENAME	SAL
106	DAVIS	3000
7		
107	ADAMS	2500
7		

```
SQL> UPDATE emp_dept7 SET SAL=SAL+300;
```

```
2 rows updated.
```

```
SQL> SELECT * FROM emp_dept7;
```

EMPNO	ENAME	SAL
106	DAVIS	3300
7		
107	ADAMS	2800
7		

--Cannot remove a row if the view contains the following:

--Group functions

--A GROUP BY clause

--The DISTINCT keyword

--The pseudocolumn ROWNUM keyword

-- 1. View using Group Functions

```
SQL> CREATE OR REPLACE VIEW emp_salary_stats AS  
SELECT
```

```
    deptno,  
    COUNT(*) AS total_employees,  
    AVG(sal) AS avg_salary,  
    MAX(sal) AS max_salary,  
    MIN(sal) AS min_salary
```

```
FROM emp
```

```
GROUP BY deptno;
```

```
  2    3    4    5    6    7    8    9
```

```
View created.
```

```
SQL> SELECT * FROM emp_salary_stats;
```

DEPTNO	TOTAL_EMPLOYEES	AVG_SALARY	MAX_SALARY	MIN_SALARY
1	1	1200	1200	1200
3	2	1425	1600	1250
2	2	2137.5	2975	1300
7	2	3050	3300	2800
	1	2000	2000	2000

```
SQL> DELETE FROM emp_salary_stats WHERE deptno=1;
```

```
\DELETE FROM emp_salary_stats WHERE deptno=1
```

```
*
```

```
ERROR at line 1:
```

```
ORA-01732: data manipulation operation not legal on this view
```

--2. View using GROUP BY Clause

```
SQL> CREATE OR REPLACE VIEW emp_count_per_job AS
SELECT
    job,
    COUNT(*) AS num_employees
FROM emp
GROUP BY job;
  2    3    4    5    6
View created.
```

```
SQL> SELECT * FROM emp_count_per_job;
```

JOB	NUM_EMPLOYEES
CLERK	2
SALESMAN	2
MANAGER	1
ANALYST	1
DEVELOPER	1
	1

6 rows selected.

```
SQL> DELETE FROM emp_count_per_job WHERE job='CLERK';
DELETE FROM emp_count_per_job WHERE job='CLERK'
      *
```

ERROR at line 1:

ORA-01732: data manipulation operation not legal on this view

--3. View using DISTINCT Keyword

```
SQL> CREATE OR REPLACE VIEW distinct_jobs AS
SELECT DISTINCT job
FROM emp;
      2      3
View created.

SQL> SELECT * FROM distinct_jobs;

JOB
-----
CLERK
SALESMAN
MANAGER
ANALYST
DEVELOPER

6 rows selected.

SQL> DELETE FROM distinct_jobs WHERE job='CLERK';
DELETE FROM distinct_jobs WHERE job='CLERK'
      *
ERROR at line 1:
ORA-01732: data manipulation operation not legal on this view
```

--4. View using ROWNUM Pseudocolumn

```
SQL> CREATE OR REPLACE VIEW top5_earners AS
SELECT empno, ename, sal
FROM (
    SELECT empno, ename, sal
    FROM emp
    ORDER BY sal DESC
)
WHERE ROWNUM <= 5;
```

2 3 4 5 6 7 8
View created.

```
SQL> SQL> SELECT * FROM top5_earners;
```

EMPNO	ENAME	SAL
106	DAVIS	3300
104	JONES	2975
107	ADAMS	2800
9999	KO KO	2000
102	ALLEN	1600

```
SQL> DELETE FROM top5_earners WHERE empno=106;
DELETE FROM top5_earners WHERE empno=106
*
```

ERROR at line 1:

ORA-01732: data manipulation operation not legal on this view

```
--CREATE [OR REPLACE] [FORCE|NOFORCE] VIEW view
--[(alias[, alias]...)]
--AS subquery
--[WITH CHECK OPTION [CONSTRAINT constraint]]
--[WITH READ ONLY [CONSTRAINT constraint]];
```

```
SQL> CREATE OR REPLACE VIEW empvu7
AS SELECT *
FROM emp
WHERE deptno = 7
WITH CHECK OPTION CONSTRAINT empvu7_ck;
      2      3      4      5
View created.
```

```
SQL> SELECT * FROM empvu7;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
106	DAVIS	ANALYST	104	18-JUN-23	3300	
107	ADAMS	DEVELOPER	106	05-JUL-23	2800	

```
SQL> INSERT INTO emp (empno,ename, job, mgr, hiredate, sal, comm, deptno)
VALUES (111, 'MOE MOE', 'DEVELOPER', 106, TO_DATE('2024-07-05','YYYY-MM-DD')
, 2500, NULL, 7);
      2
1 row created.
```

```
SQL> INSERT INTO empvu7 (empno, ename, job, mgr, hiredate, sal, comm, deptno
)
VALUES (222, 'SOE SOE', 'DEVELOPER', 106, TO_DATE('2025-01-01','YYYY-MM-DD')
, 2500, NULL,1);
      2 INSERT INTO empvu7 (empno, ename, job, mgr, hiredate, sal, comm, deptno
)
      *
ERROR at line 1:
ORA-01402: view WITH CHECK OPTION where-clause violation
```

```
SQL> CREATE OR REPLACE VIEW empvu7
AS SELECT *
FROM emp
WHERE deptno = 7;
      2      3      4
View created.
```

```
SQL> SELECT *
FROM emp;
      2
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM
101	SMITH	CLERK		15-JAN-23	1200	
102	ALLEN	SALESMAN	101	10-FEB-23	1600	300
103	WARD	SALESMAN	101	01-MAR-23	1250	500
104	JONES	MANAGER		20-APR-23	2975	

```

SQL> CREATE FORCE VIEW divinfo
AS
SELECT *
FROM divisions;
   2       3       4
Warning: View created with compilation errors.

SQL> SELECT *
FROM divinfo;
   2 FROM divinfo
      *
ERROR at line 2:
ORA-04063: view "SYSTEM.DIVINFO" has errors

```

```

SQL> CREATE TABLE divisions
(id number(5),
description varchar(20)
);
   2       3       4
Table created.

SQL> INSERT INTO divisions VALUES(1, 'aaa');

1 row created.

SQL> SELECT * FROM divinfo;

      ID DESCRIPTION
-----
      1 aaa

```

```

SQL> drop table divisions;

Table dropped.

```

```
SQL> CREATE OR REPLACE VIEW empvu10
AS SELECT *
FROM emp
WITH READ ONLY;
      2      3      4
View created.
```

```
SQL> SELECT *
FROM empvu10;
      2
```

EMPNO	ENAME				
JOB		MGR	HIREDATE	SAL	COMM
DEPTNO					

101	SMITH				
CLERK			15-JAN-23	1200	
1					
102	ALLEN				
SALESMAN		101	10-FEB-23	1600	300
3					

EMPNO	ENAME				
JOB		MGR	HIREDATE	SAL	COMM
DEPTNO					

103	WARD				
SALESMAN		101	01-MAR-23	1250	500
3					

```
SQL> INSERT INTO empvu10 (empno, ename, job, mgr, hiredate, sal, comm, deptn
o)
VALUES (333, 'SOE SOE', 'DEVELOPER', 106, TO_DATE('2025-01-01','YYYY-MM-DD')
, 2500, NULL,1);
      2  INSERT INTO empvu10 (empno, ename, job, mgr, hiredate, sal, comm, deptn
o)
```

*

```
ERROR at line 1:
ORA-42399: cannot perform a DML operation on a read-only view
```

-- NOFORCE

```
SQL> CREATE OR REPLACE NOFORCE VIEW myview
AS SELECT *
FROM mytemp;
  2      3 FROM mytemp
      *
ERROR at line 3:
ORA-00942: table or view does not exist
```

```
SQL> CREATE OR REPLACE VIEW myview
AS SELECT *
FROM mytemp;
  2      3 FROM mytemp
      *
ERROR at line 3:
ORA-00942: table or view does not exist
```

```
SQL> CREATE OR REPLACE FORCE VIEW myview
AS SELECT *
FROM mytemp;
  2      3
Warning: View created with compilation errors.
```

```
CREATE TABLE mytemp
(id number(5),
description varchar(20)
);
```

```
INSERT INTO mytemp VALUES(1,'no force test-1');
```

```
INSERT INTO mytemp VALUES(2,'no force test-2');
```

```
INSERT INTO mytemp VALUES(3,'no force test-3');
```

```
DROP VIEW empvu10;
```